



ADNAN SHAMIM

AGENTIC AI GOVERNANCE

INTERNAL AUDIT ECOSYSTEM™



TWO COMPLEMENTARY FRAMEWORKS. TWO DISTINCT PURPOSES. ONE ACCOUNTABILITY ARCHITECTURE.

STRATEGY DEFINES WHAT SHOULD HAPPEN.
ASSURANCE INDEPENDENTLY EVALUATES WHETHER IT OPERATED AS INTENDED—
AND WHETHER SUFFICIENT **EVIDENCE** SUPPORTS THAT CONCLUSION.

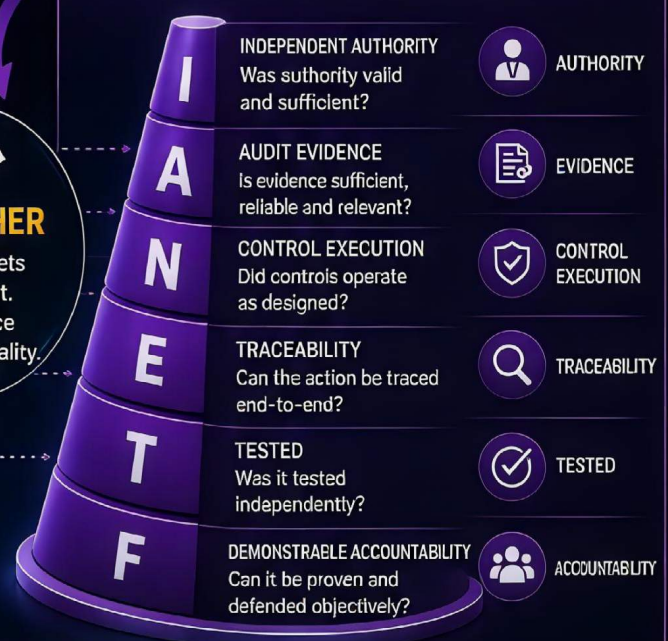
THE FUTURIST'S STRATEGY CONE™

Structures Agentic AI strategy, decision boundaries, operating models, execution, and human accountability.

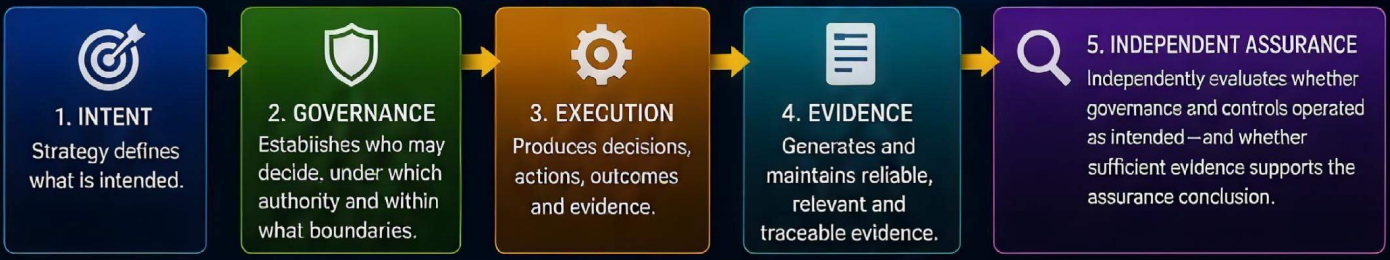


THE I-A-N-E-T-F CONE FRAMEWORK™

Structures independent assurance around authority, evidence, control execution, traceability, and demonstrable accountability.



THE DISTINCTION IS FUNDAMENTAL



WHAT THIS ECOSYSTEM ENSURES

- ✓ Clear separation of Strategy (Management) and Assurance (Internal Audit).
- ✓ Defined decision boundaries and human oversight.
- ✓ End-to-end traceability and immutable evidence.
- ✓ Independent evaluation, testing, and assurance.
- ✓ Demonstrable and defensible accountability.



WHAT THIS ECOSYSTEM DOES NOT DO

- ✗ Does not replace management accountability.
- ✗ Does not replace governance, risk management, internal controls, or compliance functions.
- ✗ Does not guarantee outcomes or eliminate risk.
- ✗ Does not limit professional judgment or audit independence.



THE OBJECTIVE IS NOT TO ASSUME CONTROL.
IT IS TO DEFINE IT, GOVERN IT, OPERATE IT,
EVIDENCE IT, TEST IT, AND INDEPENDENTLY ASSURE IT.



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Driving Responsible Agentic AI Through Strategy, Governance and Independent Assurance.



THE I-A-N-E-T-F™ CONE FRAMEWORK v2.0

Enhanced AI Assurance & Control Effectiveness Architecture

**From Documented Controls to
Demonstrable, Evidence-Bounded
Assurance**

**An Internal Audit-oriented assurance
architecture for evaluating dynamic,
AI-assisted, and Agentic AI control
environments through control-universe
completeness, control-state reconstruction,
evidence integrity, change analysis,
independent validation, and
evidence-bounded conclusions.**

1. PURPOSE & POSITIONING

The I-A-N-E-T-F™ Cone Framework v2.0 establishes a disciplined method for determining whether an assurance conclusion is supported by sufficient, reliable, relevant, and appropriately evaluated evidence within the defined objective, criteria, scope, population, and relevant period.

It is an:

Enhanced AI Assurance & Control Effectiveness Framework.

It is **not presented as an ISA/ PCAOB-required checklist**, nor does it represent every element of the framework as a verbatim requirement of any single professional standard.

It does not replace:

- management accountability;
- governance responsibilities;
- risk management;
- applicable professional standards;
- regulatory requirements;
- firm methodology;
- ethical or independence requirements; or
- engagement-specific professional judgment.

Its purpose is to provide an enhanced assurance architecture for determining whether the evidence supports the conclusion being made.

2. THE CORE I-A-N-E-T-F™ ARCHITECTURE

To preserve the framework's identity and continuity, v2.0 retains the six core pillars:

I – INTENT

Define What Should Be Permitted

Establish:

- purpose;
- objectives;
- authorized outcomes;
- prohibited outcomes;
- decision boundaries;
- operating constraints;
- risk tolerances;
- escalation conditions.

Core question:

What is the system, control, or agent intended and authorized to do—and what is it prohibited from doing?

A — AUTHORITY

Establish Who or What May Act

Determine:

- ownership;
- accountability;
- authorization;
- delegation;
- permissions;
- decision rights;
- approval thresholds;
- human oversight;
- agent-to-agent authority.

Core question:

Who—or what—has authority to make or execute the relevant decision?

N – ENFORCEMENT

Determine Whether Boundaries Actually Hold

Evaluate:

- technical controls;
- access restrictions;
- policy enforcement;
- segregation;
- preventive controls;
- detective controls;
- decision boundaries;
- tool restrictions;
- execution constraints;
- monitoring;
- exception handling.

Core question:

Are the approved boundaries technically and operationally enforceable?

E – TESTING

Demonstrate That Controls Operate

Determine whether controls:

- operated as required;
- operated within authorized boundaries;
- operated during the relevant period;
- operated at the required frequency;
- operated consistently;
- responded appropriately to exceptions;
- remained effective following relevant changes.

Core question:

Can operating effectiveness be demonstrated rather than merely asserted?

T – TRACEABILITY & EVIDENCE

Make the Control and Its Operation Reconstructable

Evaluate:

- evidence provenance;
- lineage;
- completeness;
- integrity;
- timing;
- population coverage;
- system logs;
- configuration history;
- decision records;
- intervention records;
- material outputs;
- change records.

Core question:

Can the material control activity, decision, change, and evidence chain be reconstructed?

F — FORENSICS & REVOCABILITY

Prove the Environment Can Respond When Controls Fail

Evaluate:

- anomaly detection;
- incident response;
- forensic reconstruction;
- containment;
- authority revocation;
- credential invalidation;
- session termination;
- tool blocking;
- downstream access;
- delegated-agent authority;
- cessation evidence.

Core question:

When authority must be withdrawn, can the organization demonstrate that consequential authority was actually removed?

3. v2.0 ASSURANCE EXECUTION CYCLE

The six I-A-N-E-T-F™ pillars are operationalized through the following **v2.0 Assurance Execution Cycle**:

IDENTIFY 'n ASSESS 'n NAVIGATE 'n
EVALUATE 'n TEST 'n FIND

This is the **execution methodology**, not a redefinition of the I-A-N-E-T-F™ acronym.

4. IDENTIFY

Establish the Assurance Universe

Identify:

- assurance objective;
- scope and exclusions;
- applicable criteria;
- relevant risks;
- control objectives;
- authoritative control sources;
- control universe;
- relevant period;
- populations;
- evidence requirements;
- system boundaries;
- AI/agent boundaries;

- decision boundaries;
- permissions;
- tools;
- integrations;
- delegated agents;
- technical guardrails;
- monitoring;
- revocation mechanisms.

Governing principle

A control repository is evidence of representation—not automatic proof of control-universe completeness.

The relevant universe must be established against the defined assurance objective, criteria, scope, period, risks, systems, processes, controls, and relevant authoritative sources.

5. ASSESS

Establish What Should Exist

Assess:

- control design;
- authority;
- ownership;
- risk relevance;
- expected operating conditions;
- implementation requirements;
- evidence requirements;

- materiality;
- dependencies;
- AI/model dependencies;
- permissions;
- change requirements;
- failure conditions;
- revocation requirements.

Three-State Model

Every material control should be considered across:

**EXPECTED STATE 'n IMPLEMENTED
STATE 'n OBSERVED STATE**

Expected State

What should exist?

Implemented State

How is it actually configured or deployed?

Observed State

What does reliable evidence demonstrate actually existed or occurred?

**Expected state $\neq$ Implemented state $\neq$
Observed operation**

6. NAVIGATE

Trace the Control Into Reality

Navigate from:

Requirement

,

Risk

,

Control

,

System

,

Configuration

,

Agent

,

Decision Boundary

,

Execution

,

Evidence

Determine:

- where the control actually resides;
- whether it is documented or technically embedded;
- who or what operates it;
- what permissions enforce it;
- what systems depend on it;
- what evidence it generates;
- how exceptions are handled;
- whether bypass is possible;
- whether authority can be delegated;
- whether consequential actions can occur outside approved boundaries.

Agentic AI Control Path

Authority

,

Permissions

,

Decision Boundary

,

Execution

,

Monitoring

,

Revocation

,

Evidence

7. EVALUATE

Determine Whether the Evidence Can Support the Assertion

Evaluate:

- relevance;
- reliability;
- sufficiency;
- appropriateness;
- provenance;
- completeness;
- accuracy;

- precision;
- timing;
- integrity;
- lineage;
- population coverage;
- retention;
- alteration risk;
- independence;
- limitations.

Non-Negotiable Principle

AI-generated or AI-processed information is not self-authenticating evidence.

Likewise:

System-generated evidence is not automatically reliable merely because it was produced by a controlled system.

The evidence-generating mechanism itself may require evaluation where its reliability is relevant to the assurance conclusion.

AI Evidence Reliability Gate

SOURCE

,

INTEGRITY

,

COMPLETENESS

,

PROCESSING

,

OUTPUT

,

VALIDATION

,

PURPOSE

,

CONCLUSION

8. THE THREE COMPLETENESS GATES

Gate 1 – Universe Completeness

Have we identified the controls relevant to the defined assurance objective?

A complete repository does not prove operating effectiveness.

Gate 2 – State Completeness

Have we sufficiently captured the relevant control states for the period under examination?

A current state does not prove historical operation.

Gate 3 – Evidence Completeness

Do we possess sufficient appropriate evidence to support the specific assertion?

A large evidence population does not prove universe completeness.

9. TEST

Demonstrate Operating Effectiveness

Test whether the control:

- operated as designed or otherwise as required;
- operated within its authorized boundary;
- operated during the relevant period;
- operated at the required frequency;
- operated consistently;
- prevented or detected relevant unauthorized activity;
- generated sufficiently reliable evidence;
- responded appropriately to exceptions;
- remained effective following relevant changes.

AI / Agentic AI Testing

Where relevant, test:

- model/version;
- configuration;
- material instructions;
- permissions;
- credentials;
- tool access;
- data lineage;
- output validation;
- human oversight;

- decision thresholds;
- exception handling;
- agent-to-agent interactions;
- downstream actions;
- monitoring;
- change controls;
- drift;
- revocation.

No operating-effectiveness conclusion from design documentation alone.

No historical-operation conclusion from a current snapshot alone.

10. FIND

Form the Evidence-Bounded Assurance Conclusion

“Find” means more than identifying exceptions.

It means determining:

- what the evidence demonstrates;
- what it does not demonstrate;
- whether exceptions are significant;
- whether deficiencies affect the control objective;
- whether broader populations or periods may be affected;
- whether additional procedures are necessary;
- whether assurance limitations exist;
- **the conclusion the evidence legitimately supports.**

Fundamental Rule

The assurance conclusion must be no stronger—and no weaker—than the evidence, criteria, scope, period, and professional evaluation legitimately support.

11. HISTORICAL STATE & DELTA ASSURANCE

v2.0 explicitly distinguishes:

CURRENT STATE $\neq$ HISTORICAL STATE

For material controls, determine:

1. What should have existed?
2. What was implemented?
3. What was observed?
4. What existed during the relevant period?
5. What changed?

6. When did it change?
7. Was the change authorized?
8. Was the change appropriately implemented?
9. Was the resulting state within the authorized boundary?
10. Did the change affect the control objective or risk?
11. What evidence supports the conclusion?

Delta Principle

A detected delta is not automatically a control failure.

And:

No detected delta is not automatically proof that no change occurred.

The strength of a no-change conclusion depends upon the completeness, reliability, coverage, and capability of the detection process.

12. AGENTIC AI ASSURANCE CONTROL LAYER

For Agentic AI, v2.0 evaluates the complete **authority-to-action chain**.

1. AUTHORITY

Who authorized the agent and what authority was granted?

2. PERMISSIONS

What can the agent access, modify, execute, or communicate?

3. DECISION BOUNDARIES

What decisions may the agent make autonomously?

4. EXECUTION

What did the agent actually cause to occur?

5. MONITORING

What detects abnormal, unauthorized, or materially changed behavior?

6. REVOCATION

Can authority actually be removed?

7. EVIDENCE

Can the material action chain be reconstructed?

13. EFFECTIVE REVOCATION TEST

A documented “kill switch” is insufficient by itself.

The assurance question is:

Can Internal Audit demonstrate that the agent lost the technical ability to perform the prohibited consequential action?

Where relevant, testing should address:

- authorization revocation;
- credential invalidation;
- session termination;
- permission removal;
- tool blocking;
- downstream access;
- delegated-agent authority;
- queued actions;
- scheduled actions;
- persistent credentials;
- subsequent prohibited-action attempts;
- evidence of effective cessation.

Core Principle

**STOP COMMAND $\neq$ EFFECTIVE
REVOCATION**

**EFFECTIVE REVOCATION =
DEMONSTRATED LOSS OF
CONSEQUENTIAL AUTHORITY**

14. AI EVIDENCE RELIABILITY GATE

Before relying materially on AI-generated or
AI-processed information:

SOURCE 'n INTEGRITY 'n COMPLETENESS
'n PROCESSING
 'n OUTPUT 'n VALIDATION 'n
PURPOSE 'n CONCLUSION

Do not treat AI output as reliable merely because:

- the model is sophisticated;
- the provider is reputable;
- the system supplies a confidence score;
- the output appears logical;
- the output is professionally formatted;
- another AI system reviewed it;
- a human approved it.

15. THE FIVE-GATE RELIANCE TEST

Before material reliance:

GATE 1 – AUTHORIZED

Was the activity authorized for the defined purpose?

GATE 2 – CONTROLLED

Were relevant controls implemented and operating within authorized boundaries?

GATE 3 – TRACEABLE

Can the material activity and evidence be reconstructed?

GATE 4 – INDEPENDENTLY VALIDATED

Was the evidence critically evaluated rather than merely accepted?

GATE 5 – SUFFICIENTLY SUPPORTED

Does sufficient appropriate evidence support the conclusion?

If a gate fails, material reliance must be restricted, suspended, supplemented, or otherwise evaluated under the applicable professional requirements.

16. I-A-N-E-T-F™ v2.0

ASSURANCE

ARCHITECTURE

	ASSURANCE
OBJECTIVE	,
	SCOPE + CRITERIA +
PERIOD	,
	CONTROL UNIVERSE
	,
	EXPECTED STATE
	,
	IMPLEMENTED STATE
	,
	OBSERVED STATE
	,
	HISTORICAL STATE
	,
	DELTA / CHANGE
	,
	EVIDENCE VALIDATION
	,
	INDEPENDENT
VALIDATION	,
	EXCEPTION / LIMITATION
ASSESSMENT	,
	EVIDENCE-BOUNDED
CONCLUSION	

17. NON-NEGOTIABLE I-A-N-E-T-F™ v2.0 RULES

- 1. No assertion without a defined criterion.**
- 2. No control-universe claim without a defined assurance objective and scope.**
- 3. No operating-effectiveness conclusion from design documentation alone.**
- 4. No historical-operation conclusion from a current snapshot alone.**
- 5. No evidence accepted without appropriate consideration of its reliability and relevance.**
- 6. No AI-generated output treated as inherently reliable or self-authenticating evidence.**
- 7. No management assertion substituted for independent evaluation.**

- 8. No detected delta automatically classified as a control failure.**
- 9. No absence of evidence automatically treated as evidence of absence.**
- 10. No “no detected change” conclusion treated as proof of no change without adequate detection capability and coverage.**
- 11. No stop command treated as proof of effective agent revocation.**
- 12. No human sign-off treated as automatic proof of independent validation.**
- 13. No vendor certification treated as engagement-specific assurance.**
- 14. No assurance conclusion stronger than the evidence legitimately supports.**

18. WHAT v2.0 DOES NOT CLAIM

I-A-N-E-T-F™ v2.0 does not claim that:

- every undocumented activity is automatically a control deficiency;
- every change is automatically a control failure;
- every missing record proves non-operation;
- a current snapshot proves continuous historical operation;
- a control repository is automatically complete;
- no detected delta proves no change;

- system-generated evidence is automatically reliable;
- AI output automatically constitutes audit evidence;
- human review automatically establishes independent validation;
- vendor certification establishes engagement-level assurance;
- Internal Audit assumes management's responsibility;
- every framework element is expressly required by ISA, PCAOB, ISO, or another individual standard.

Instead:

I-A-N-E-T-F™ v2.0 provides a disciplined methodology for determining whether the available evidence supports the specific assurance conclusion being made.

19. FINAL ASSURANCE FORMULA

CONTROL UNIVERSE + CRITERIA +
EXPECTED STATE + IMPLEMENTED
STATE + OBSERVED STATE + HISTORICAL
STATE + EVIDENCE + TIME + DELTA +
INDEPENDENT VALIDATION =
DEFENSIBLE ASSURANCE

No individual component is sufficient by itself.

20. THE ULTIMATE I-A-N-E-T-F™ PRINCIPLES

You cannot reliably assure change against an inadequately defined universe.

You cannot establish operating effectiveness from documentation alone.

You cannot infer sustained operation from a single snapshot.

You cannot treat absence of evidence as automatic evidence of absence.

You cannot treat AI-generated information as inherently reliable.

You cannot convert an observed difference into a control conclusion without evaluating criteria, evidence, context, authorization, and risk.

And you cannot make an assurance conclusion stronger than the evidence legitimately supports.

21. THE v2.0 TRANSFORMATION

DOCUMENTED GOVERNANCE

/

DEMONSTRATED GOVERNANCE

/

DEFENSIBLE ASSURANCE

22. THE ULTIMATE QUESTION

“Can we independently demonstrate it?”

—not merely—

“Do we say it exists?”

FINAL POSITIONING

THE I-A-N-E-T-F™ CONE FRAMEWORK v2.0

**Enhanced AI Assurance & Control
Effectiveness Architecture**

**From Documented Controls to
Demonstrable, Evidence-Bounded Assurance**

An Internal Audit-oriented framework for evaluating:

- control-universe completeness;
- control state;
- evidence integrity;
- historical state;
- change;
- operating effectiveness;
- Agentic AI authority;
- enforcement;
- traceability;
- revocability; and
- evidence-bounded assurance conclusions.

This is an enhanced assurance architecture—not an ISA/PCAOB-required checklist.

It does not replace applicable professional standards, regulatory requirements, management accountability, governance, risk management, firm methodology, or professional judgment.

It is designed to make assurance conclusions rigorous, evidence-bounded, standards-aware, and resistant to common assurance, evidence, governance, change-management, and Agentic AI control loopholes.

I-A-N-E-T-F™ CORE ARCHITECTURE

**INTENT 'n AUTHORITY 'n ENFORCEMENT
'n TESTING 'n TRACEABILITY & EVIDENCE
'n FORENSICS & REVOCABILITY**

v2.0 ASSURANCE EXECUTION CYCLE

**IDENTIFY 'n ASSESS 'n NAVIGATE 'n
EVALUATE 'n TEST 'n FIND**

The result

**Declared governance 'n Demonstrated
governance 'n Defensible assurance.**

THE I-A-N-E-T-F FRAMEWORK™

1

AUDITING AGENTIC AI

BEYOND POLICIES.
BEYOND APPROVALS.
TOWARD PROOF.



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THE REAL AUDIT QUESTION

2

“

Can management demonstrate, through sufficient, reliable, relevant and useful evidence, that the agent operates within its approved purpose, authority, risk tolerance, decision boundaries, and control environment — and that those controls are designed, implemented, enforced, operating effectively, monitored, change-controlled, and capable of timely intervention throughout the agent's lifecycle?



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THE SIX PILLARS OF ASSURANCE

3



INTENT

What is the agent authorized and prohibited to do?



AUTHORITY

Who or what authorizes each material action?



ENFORCEMENT

Can the system technically prevent violations where required?



TESTING

Is the agent continuously and adversarially tested?



TRACEABILITY & EVIDENCE

Can actions and decisions be independently reconstructed?



FORENSICS & REVOCABILITY

Can we detect, stop, contain, revoke, recover and reconstruct?

I-A-N-E-T-F FRAMEWORK™

PILLAR I – INTENT

4

Bounded. Codified. Verifiable.



- ✓ Explicit purpose & prohibited actions
- ✓ Clear decision boundaries & risk tolerance
- ✓ Data, time, geographic & counterparty limits
- ✓ Version controlled & deployment integrity
- ✓ Mapped to enforceable controls

Unbounded intent = Unauditable risk.

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PILLAR A – AUTHORITY

5

Every Material Action Must Have Verifiable Authority.



- ✓ Clear authorization model (Human/AI/Hybrid)
- ✓ Verified at point of execution
- ✓ Least privilege & segregation of duties
- ✓ Real-time, step-up & context-aware approvals
- ✓ Immutable authorization evidence

No Authority. No Action.

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PILLAR N – ENFORCEMENT

6

Prevention Must Be Technical,
Not Just Procedural.



- ✓ Hard controls over soft controls
- ✓ System-level prevention (not just policies)
- ✓ Fail-safe defaults & deny-by-design
- ✓ Validated under real-world conditions
- ✓ Control effectiveness continuously verified

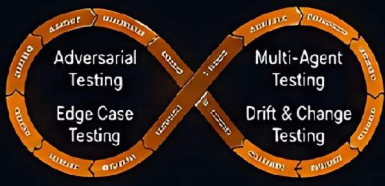
If it can't be enforced,
it can't be relied on.

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PILLAR E – TESTING

7

Continuous. Adversarial.
Multi-Dimensional.



- ✓ Automated + Human-led + Red Teaming
- ✓ Covers single & multi-agent behaviors
- ✓ Tests before, during & after changes
- ✓ Detects drift, degradation & emergent risks
- ✓ Evidence of test quality & coverage

If you don't test it,
don't trust it.

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PILLAR T – TRACEABILITY & EVIDENCE

8

If It Isn't Traceable,
It Didn't Happen.



- ✓ End-to-end trace from authorization to outcome
- ✓ Immutable, tamper-evident evidence
- ✓ Complete context (data, prompts, decisions)
- ✓ Trustworthy logs & cryptographic integrity
- ✓ Independent reconstruction always possible

Good evidence.
Better assurance.

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PILLAR F – FORENSICS & REVOCABILITY

9

Detect Fast. Stop Fast. Recover Strong.



- ✓ Detect anomalies in real-time
- ✓ Stop, contain & revoke authority
- ✓ Recover systems & restore trust
- ✓ Reconstruct events for root cause
- ✓ Notify stakeholders & regulators

Assume Failure.
Prove Resilience.

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THE BOTTOM LINE

10

Declared governance
is an assertion.

Enforced governance
is demonstrable.

Auditable governance
is demonstrable with
reliable evidence.



I-A-N-E-T-F
FRAMEWORK™

From Approval to Assurance.
From Policy to Proof.

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